

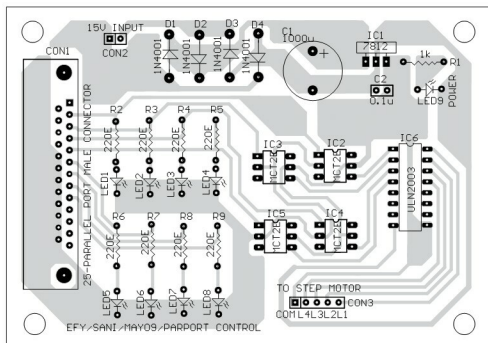


Fig. 9: Component layout for the PCB

gives the value to 'handles.dir' using 'get' command:

```
handles.dir=get(handles.  
togglebutton1, 'Value');
```

Running the program

Install Matlab 7.0.1 software in your computer system. Copy 'LPT port' folder from this month's EFY-CD to 'work' folder. Click the Matlab icon. Navigate to 'LPT port' folder from 'Current Directory' section, which is just on top of 'Command' window. To run the program, type 'data_port_controller' in 'Command' window.

The control of LED has been ex-

plained earlier. To control the stepper motor, click 'Stepper Controller' button in the program output shown in Fig. 3. You will get the program output as shown in Fig. 7. Here, you can control the stepper motor in three modes: single-step, high-torque and half-step. Also, the motor can be controlled either in forward or reverse direction. By default, the program is designed to rotate the stepper motor in reverse direction with 100ms delay. You can change these parameters as per your requirement.

Generating standalone program. To run the program without Matlab, create a standalone '.exe' file from 'M-file'

and port it to other computers using MCR (Matlab component run time). For that, configure the Matlab compiler as follows: Type 'mbuild -setup' against the prompt. Matlab will ask for a few compiler options. Select the default Matlab compiler. Then type 'mcc -m mfilename.m -o exefilename'. Here 'mfilename' must be of function 'm-file' and 'exefilename' is the filename of the standalone executable file that will be generated. This command generates a few other files along with standalone '.exe' file. Now, you can run the executable file on a PC without Matlab by copying the generated files to that computer.

Construction

Assemble this project on any general-purpose PCB or on the PCB shown here in Fig. 8. Component layout for the same is shown in Fig. 9. After mounting all the components on the PCB, fix the 25-pin male connector (CON1) on the PCB as shown in the layout. Use a 9-wire, 50cm FRC cable with 25-pin female connector at one end and male connector at the other end to connect between the board and the PC. Identify the stepper-motor wires and connect the common wire to 12V and the rest of the coils in sequence as shown in the schematic diagram.

EFY note. All the relevant files of this project have been included in this month's EFY-CD.

DATA_PORT_CONTROLLER.M

```
function varargout = data_port_  
controller(varargin)  
% DATA_PORT_CONTROLLER M-file for data_port_  
controller.fig  
%  
% DATA_PORT_CONTROLLER, by itself,  
% creates a new DATA_PORT_CONTROLLER or raises  
% the existing  
% singleton.  
%  
% H = DATA_PORT_CONTROLLER returns the  
% handle to a new DATA_PORT_CONTROLLER or the  
% handle to  
% the existing singleton.  
%  
% DATA_PORT_CONTROLLER('CALLBACK',hobj  
% ect,eventData,handles,...) calls the local  
% function named CALLBACK in DATA_  
% PORT_CONTROLLER.M with the given input  
% arguments.  
%  
% DATA_PORT_CONTROLLER('Property','Val  
% ue',...) creates a new DATA_PORT_CONTROLLER  
% or raises the  
% existing singleton. Starting from  
% the left, property value pairs are  
% applied to the GUI before data_port_  
% controller_OpeningFunction gets called.
```

```
An  
% unrecognized property name or  
% invalid value makes property application  
% stop. All inputs are passed to  
% data_port_controller_OpeningFcn via var-  
% argin.  
%  
% *See GUI Options on GUIDE's Tools  
% menu. Choose "GUI allows only one  
% instance to run (singleton)".  
% See also: GUIDE, GUIDATA, GUIHANDLES  
% Edit the above text to modify the re-  
% sponse to help data_port_controller.  
% Last Modified by GUIDE v2.5 28-Sep-2008  
% 10:23:03  
% Begin initialization code - DO NOT EDIT  
gui_Singleton = 1;  
gui_State = struct('gui_Name', mfile-  
name, ...  
'gui_Singleton', gui_  
Singleton, ...  
'gui_OpeningFcn', @  
data_port_controller_OpeningFcn, ...
```

```
'gui_OutputFcn', @  
data_port_controller_OutputFcn, ...  
'gui_LayoutFcn', []  
, ...  
'gui_Callback', []);  
if nargin & isstr(varargin{1})  
    gui_State.gui_Callback =  
    str2func(varargin{1});  
end  
if nargin  
    [varargout{1:nargout}] = gui_  
mainfcn(gui_State, varargin{:});  
else  
    gui_mainfcn(gui_State, varargin{:});  
end  
% End initialization code - DO NOT EDIT  
% --- Executes just before data_port_  
% controller is made visible.  
function data_port_controller_  
OpeningFcn(hObject, eventdata, handles,  
varargin)  
% This function has no output args, see  
% OutputFcn.  
% hObject handle to figure
```